

A High Speed Cassette Interface

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Mass Storage

One of the main advantages of a computer is the ability to easily change the function of the device by simply loading a new program. In the simplest case this is done by typing in a new set of instructions by hand. This soon gets pretty boring, even for a short BASIC program, but imagine loading the BASIC interpreter by hand each time you wanted to use it! Clearly some form of mass storage is an urgent requirement for any computing system. In the past mini-computer systems have tended to use paper tape as a (relatively) cheap storage device, but paper-tape equipment tends to be fairly noisy and expensive. Consequently the most popular mass storage medium for personal computing is magnetic tape cassettes. With care low cost audio recorders can give reliable recording and playback.

Having decided that cassette recorders are a good thing, how should the data be recorded?

The first thing to do is to convert the parallel data byte (eight bits on eight lines) to a serial form (eight bits one after the other on a single line). This can either be done synchronously, whereby a continuous stream of data is transmitted, or asynchronously, in which timing information precedes and follows each eight bits of data. For various reasons, particularly its tolerance of speed and other occasional errors, an asynchronous format is generally used with audio cassette recorders. An exception to this is the Tarbell interface, which uses synchronous recording. In general this is reported to operate reliably, but should it 'drop' a bit and get out of synchronization the rest of the record is garbled.

Waveform

The waveform for the asynchronous transmission of a byte of data is shown in fig. 1. In the idle condition (no data being transmitted) the signal is high. It goes low for one bit period to signify that data is on its way. This is followed by up to eight bits of data, least significant bit first. This may be followed by a parity bit which can be used to check the received data for errors. Finally the signal is held high for one or two bit periods to signify the end of the data. This also ensures that the receiver will see the high to low transition required to indicate a new start bit. This is all pretty complicated, but fortunately the semiconductor manufacturers have done all the work for us, and devices known as UARTs (Universal Asynchronous Receiver Transmitter) are readily available. A common pin-out is shown in fig. 2. The various control inputs allow the UART to be set up to give five to eight bits of data, odd, even or no parity and one or two stop bits. Status outputs indicate what the device is up to. The timing of the transmitter and receiver sections is controlled by two clock inputs, which must be at 16 times the data rate. The UART receiver samples the incoming data at what it expects to be the middle of each data bit. If the clock is too fast or slow the sampling point will move towards the start or finish of the data bit. Depending on the word length the clock must be accurate to about 5 to 9% for correct reception.

Recording

Now that the data has been turned into a tidy string of high and low logic levels, it must be converted to a

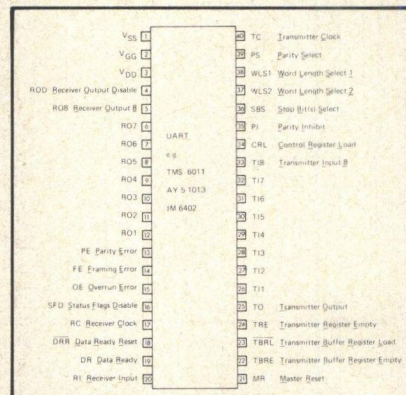


fig. 2. UART pin configuration

suitable form for recording. Direct recording of the UART output is not satisfactory owing to the wide variation of the DC level of the signal. The next simplest approach is to switch an audio tone on for a one and off for a zero. Unfortunately the automatic level controls fitted to most cheap recorders get a bit unhappy when fed with this sort of signal. Also the need to detect the tone requires a fairly long burst, and hence a rather low data rate. The next step is to use two tones, one for a one and the other for a zero. This is the basic principle of CUTS, a format sponsored by BYTE magazine to provide a standard for reliable, interchangeable cassette recording. This uses tones at 2400 and 1200 Hz. A *one* consists of eight cycles at 2400 Hz, while a *zero* consists of four cycles of 1200 Hz. These frequencies were chosen to give a standard 300 baud data rate, and to allow for easy regeneration of the 16 times clock (4800 Hz) for the UART. In theory the high redundancy (using several flux transitions to convey each bit of information) gives a highly reliable recording, but in practice few receiver circuits make full use of the redundancy.

CUTS is still slow, an 8K byte program taking about four minutes to load in binary format, or ten minutes in a hexadecimal format. In

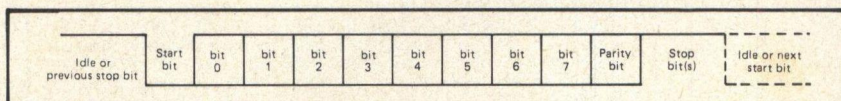


fig. 1. UART data format

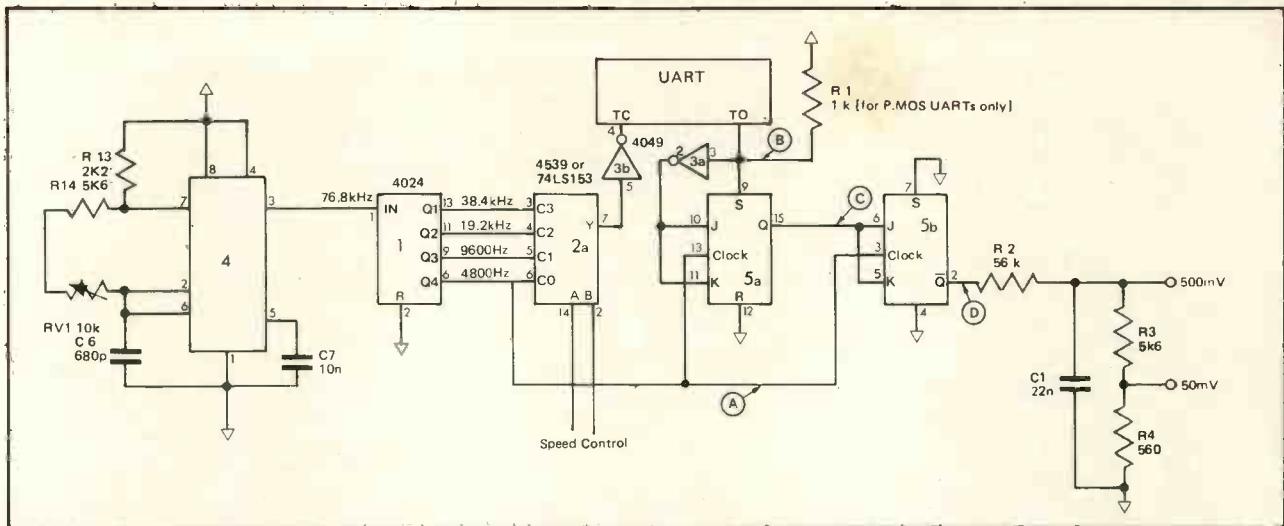


fig. 3. Transmitter Circuit.

order to speed up the recording the number of cycles of 1200 and 2400 Hz can be reduced by successive factors of two to obtain 600, 1200 and 2400 baud. At the maximum speed a one consists of 1 cycle of 2400 Hz, and a zero is half a cycle of 1200 Hz.

This interface can operate at all of these speeds from standard CUTS at 300 baud to 'high speed CUTS' at 2400 baud.

divide the 4800 Hz clock input by two or four, giving the required CUTS format. This is put through a simple low pass filter to remove the fast edges of the square wave. For higher speed recording the multiplexer selects a faster clock for the UART, which shortens the duration of each bit. The transmitter waveforms at 2400 baud are shown in fig. 5.

ing levels in the absence of an input signal. The two exclusive-or gates (IC10b and c) then form a transition detector, which gives a negative-going output pulse each time the input changes state. These pulses are inverted and used to trigger a 'digital monostable' (IC9a). Providing this monostable is triggered (i.e. the counter is reset) before 12 clock pulses have been counted, the input to the D-type flip-flops (IC7) remains high, and the inverted output from the second flip-flop (IC7b) remains high. This will be the case whenever a 2400 Hz input signal is being received. With a 1200 Hz input signal the counter will reach 12 about 3/4 of the way through the half cycle, taking the output of the NAND gate (IC6d) low. The next pulse from the transition detector takes the output from IC7a and IC3d low. Two more 2400 Hz pulses from the transition detector are required to send the inverted output of IC7b high again.

This serves to extend the duration of the low output to the correct time.

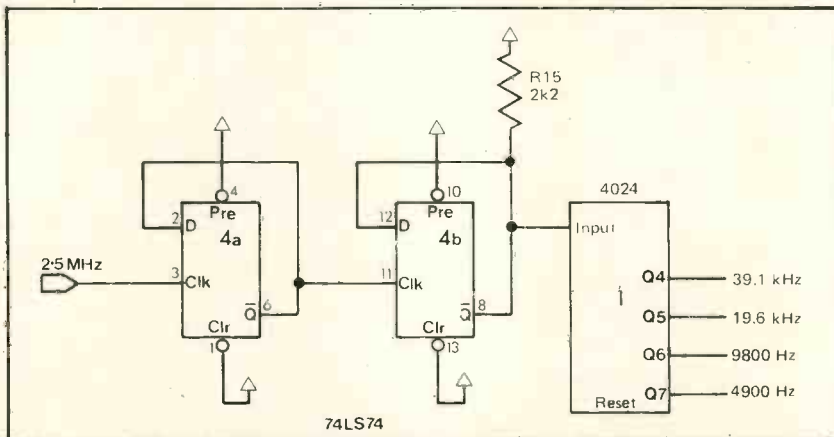


fig. 4. Deriving Master Oscillator from 2.5 MHz

The Transmitter

The transmitter circuitry is shown in fig. 3. The timing of the entire interface is controlled by a master oscillator. In fig. 3, this is shown as a 555 operating at 76.8 kHz, but it can be derived from any stable source, a crystal being ideal. In particular a Z80 system running at 2.5 MHz can give a master clock within 1% of the correct frequency. A suitable circuit is shown in fig. 4.

The 16 times clock for the UART is selected by the multiplexer, IC2a, which is controlled by the two speed control inputs. For standard CUTS the frequency of this clock is 4800 Hz, giving a transmitter output at 300 baud. This output controls whether the two J-K flip-flops, IC5,

The Receiver

Putting the data onto tape is fairly easy, the hard part is getting it back again. The first thing to do is to get rid of as much extraneous noise as possible (particularly mains hum), and then convert the signal to a nice clean square wave. This is achieved by IC3c and IC10a (fig. 6) in conjunction with the associated circuitry. C2 and RV1 form a high pass filter which attenuates low frequency signals such as mains hum. The signal is then amplified by a CMOS inverter IC3c, used in the analog mode (use the unbuffered version if possible). The resultant signal is squared by the schmitt trigger, IC10a. The DC conditions can be adjusted with RV1 such that the input to the schmitt trigger is in between the two switch-

Feeding the UART receiver clock

Having extracted the data, we must now obtain a signal at 16 times the receiver bit rate to feed to the UART receiver clock. In a perfect world this could be the same as the transmitter clock. This would probably work if the same recorder were always used for both recording and playback, but in general low cost cassette recorders tend not to have very accurate tape speeds. Consequently this approach can give difficulties in playing back tapes recorded on other machines. Fortunately, the signal from the recorder contains sufficient informa-

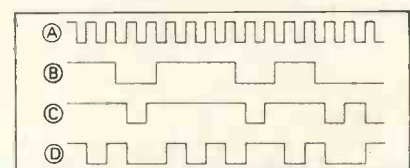


fig. 5. Transmitter Waveforms

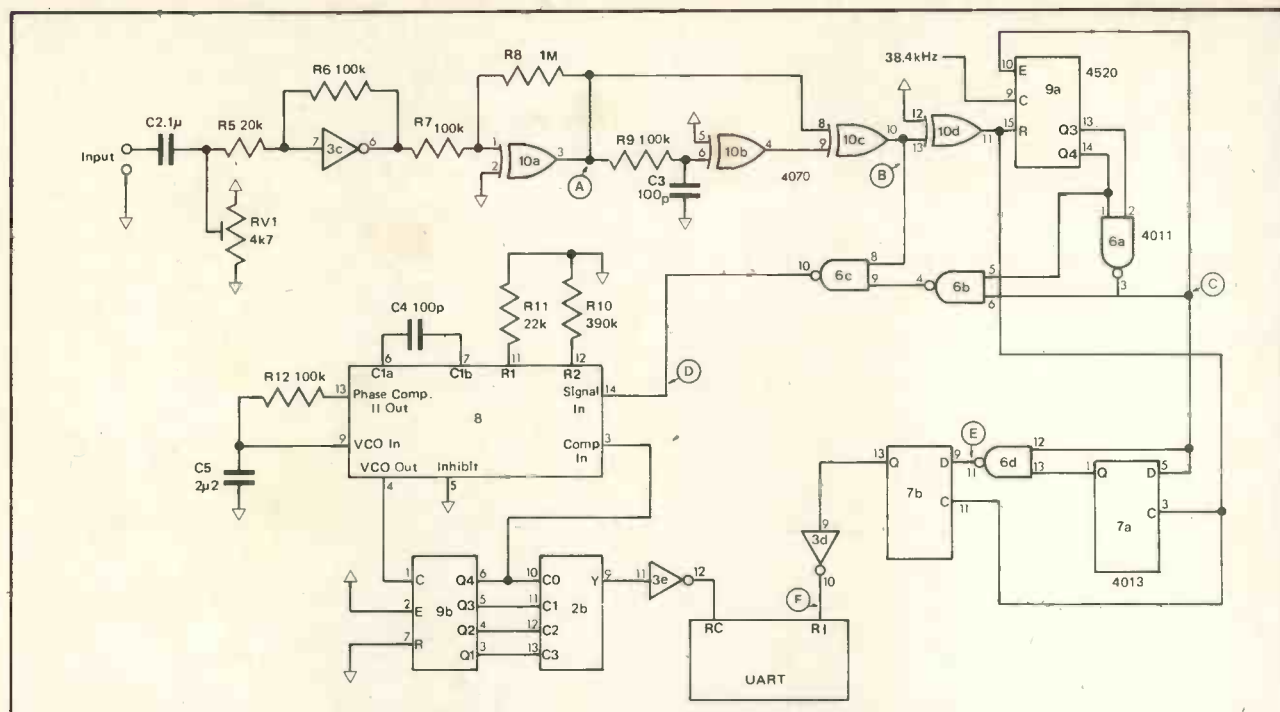


fig. 6. Receiver Circuit

ation to reconstruct the clock frequency. When a 2400 Hz signal is being received the transition detector will give an output at 4800 Hz. With a 1200 Hz signal the transition detector will give an output of only 2400 Hz, so an extra pulse must be found from somewhere. Such a pulse is available from the counter (IC 9a) which is acting as a digital monostable. Nominally this counter will reach eight exactly half way through the half cycle. This is all we need for CUTS, but at the higher speeds we need a higher frequency.

Once again, the semi-conductor makers come to our aid with a device known as a phase locked loop (PLL). For our purposes this can be considered as a device which will adjust its output frequency to keep two input frequencies the same. Obviously one of the input frequencies must be derived from the output to keep the loop stable, but a divide by N counter can be fitted between the input and the output, whereupon the output frequency will be N times the signal input frequency. Thus the phase locked loop can, in effect, multiply an input frequency by any whole number. In this circuit the phase locked loop is designed to operate at 76.8 kHz and the required signals of 38.4, 19.2, 9.6 and 4.8 kHz can be obtained from the feedback divider, IC9b. The correct frequency is selected by the multiplexer, IC2b, in the same way as for the transmitter, and fed to the 16-times clock of the UART. The inverting buffer provides a higher drive capability for use with P-MOS UARTs. The waveforms at various points in the circuit are shown in fig. 7.

Connecting the Interface to Your Computer

This interface can be used with a ready made board which already contains a UART. All that is needed is to connect the transmitter and receiver clocks and the serial in and out from the UART as shown in figs. 3 and 6. The two speed control bits can be connected to an existing output port if software selectable speeds are required, or a 2 pole, 4 way switch can be used for manual speed control, as shown in fig. 8.

If the UART is also being added to the system, several options are available, depending on how much control you want the computer to have over the UART. In the most basic system the computer must be able to read and write data for the UART, and also read the UART status, particularly DR (Data Ready). The various control inputs such as the number of stop bits and word length must be hard-wired to logic 1 or 0, and the UART allowed to see them by wiring CRL (Control Register Load) to logic 1.

At times, however, it may be useful to be able to change these options, and this can be done by addressing the control register as an output port. If one of the control inputs, such as

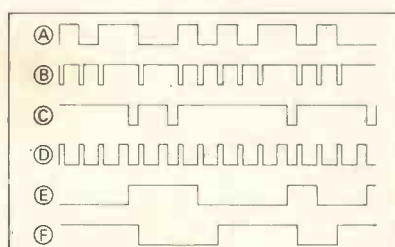


fig. 7. Receiver Waveforms

WLS2, is hard-wired, this leaves four spare lines on the control port, and these can be latched by a 74(LS)175 or 74(LS)75 to be used to select the recording speed, and to switch two reed relays for recorder control. This allows two separate recorders to be used, one for recording and one for playback. With a bit of thought on the software, this will permit large blocks of text, such as assembler source listings, to be handled without needing massive amounts of memory. An example of this approach can be seen in Patrick Sutton's article in PCW Vol. 1, No. 2.

One way of connecting the UART is shown in fig. 9. This uses the E78 bus conventions, but only minor changes should be needed to adapt the circuit to other bus structures. The 74LS138 is actually decoding four read and write addresses, and a second UART can be fitted with no extra decoding. This could be used to provide a second cassette interface, or to give a V24 interface for a VDU, Teletype or similar device.

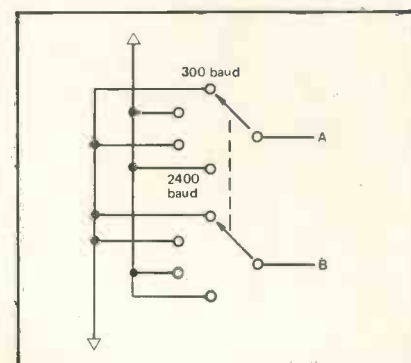


fig. 8.
Switch Selection of Operating Speed

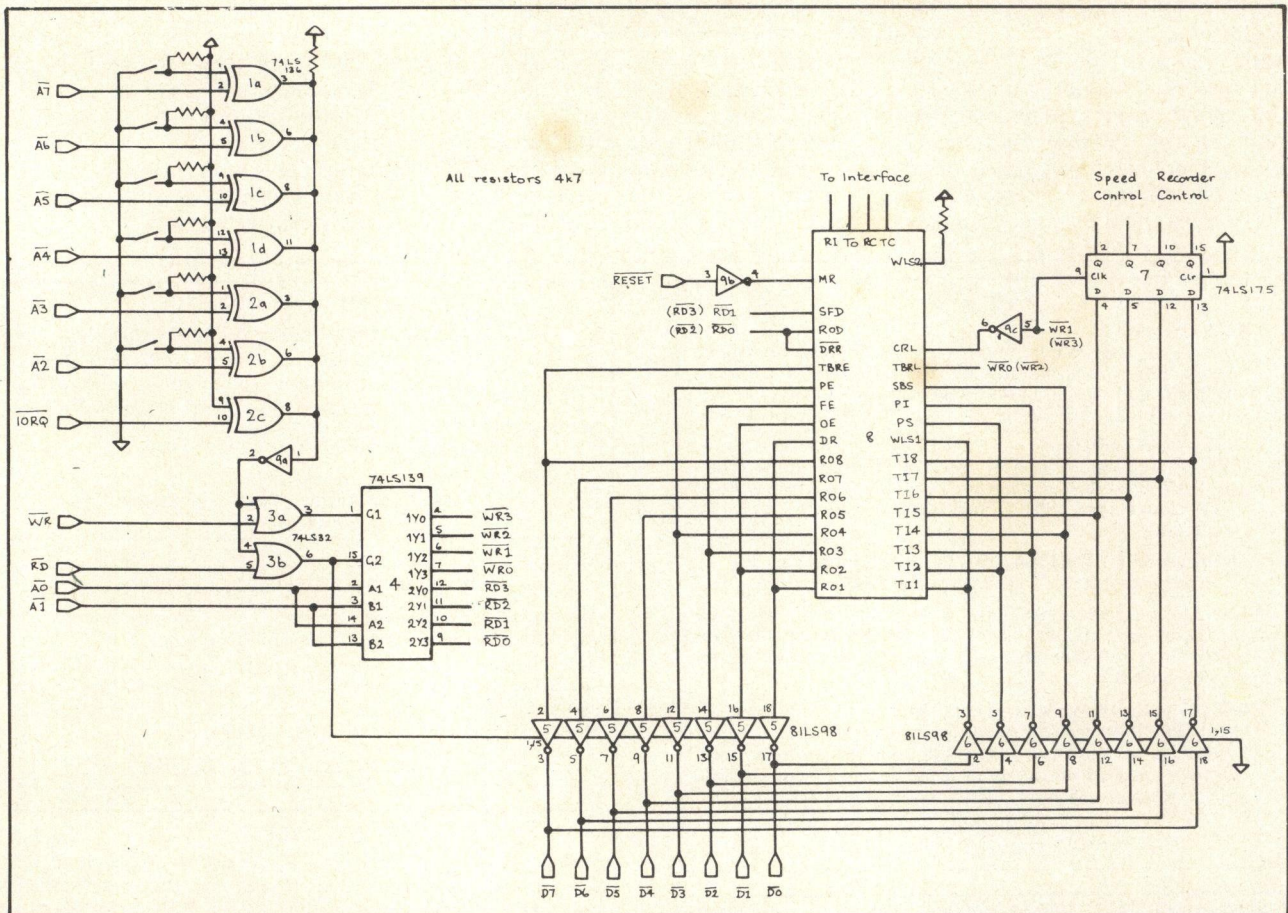


Fig. 9

Does It Work?

As far as we can tell this system is very reliable, even at 2400 baud. The few errors that have occurred have generally been associated with particular patches of tape, especially in the first foot or so of a cassette. This system does record at a very high density, and high quality tape should be used. It is inadvisable to use tapes longer than C60, and cassettes should be carefully stored away from dusty atmospheres etc. The cassette recorder heads should be kept clean.

Because it extracts its clock from the incoming data this circuit is very tolerant of speed variations. With the prototype the master oscillator frequency could be adjusted from 30% below to 13% above its nominal value. While a recording was being made the frequency was swept from one extreme to the other. The frequency was reset to its nominal value, and the tape was read with no errors.

Construction

Providing the normal precautions are taken to avoid discharging static electricity into the CMOS devices, there should be no problems in constructing the circuit.

The only adjustments required are to set the master oscillator frequency and to set the DC input conditions with RV1. The latter is just a matter

of looking at the output of IC10a with a meter or logic probe, and "twiddling" RV1 to about half-way between the two points at which it changes state. The gain of the input should be adequate for most cassette recorders, if required it can be adjusted by changing R6. The gain is simply $R6/R5$ (e.g. the gain as shown is five). If a crystal oscillator is used there are no further adjustments required to the circuit, although the best tone and volume settings of the recorder should be found by trial and error. If an RC oscillator is used this must be adjusted to give an idle output frequency of 2400Hz. This should be done as accurately as possible (for the sake of friends with less speed tolerant CUTS circuits), though tuning slightly sharp of top D on a piano should give reasonable results.

The circuit, being CMOS, should not be sensitive to layout, and can be constructed on Veroboard or by wire-wrapping. For lazy constructors a small circuit board carrying all the circuit except for the UART can be obtained from **Newbear**. Detailed instructions on connecting this board to the Nascom microcomputer are also available. An E78 circuit board is also being prepared, and will include this interface, a second UART with a standard V24 interface, and space for wiring additional input and

output ports. This again should be available from **Newbear**.

For The Experimenter

Just as any program can be improved, so this circuit is by no means the ultimate cassette interface. One intriguing possibility is that the output from the phase locked loop of the receiver could be used to drive the digital monostable. This would allow the receiver to track an even wider range of frequencies. Unfortunately it also gives a system with several stable states, and it doesn't work with the circuit as it stands.

Another possibility is that of running at even higher data rates by increasing the clock oscillator frequency and adjusting the phase locked loop. The prototype has operated for a short time at 5000 baud without errors, but you do this at your own risk!

Because of the inevitable variations between components, the resistor and capacitor values for the phase locked loop have been selected to give a fairly wide range over which the loop will lock. In this context it should be noted that all 4046s do not seem to be equal. The circuit was designed using a Motorola equation, and works using Motorola chips. Occasional chips from other makers may give trouble, which can be overcome by adjusting the range over which the loop will lock. First adjust

the low frequency to 50kHz. Temporarily connect a 10 k Ω resistor from pin 9 of the PLL to ground. Measure the output frequency (if you don't have a frequency meter you can still measure the frequency by connecting the output of the feedback counter to a spare bit of an input port, and letting your com-

puter do the work). The ideal resistor value is then the current value of R10 times the current frequency (in kHz) divided by 50. The upper frequency may be adjusted in a similar way to 110 kHz. Move the end of the 10 k Ω resistor from ground to +5V, measure the output frequency and adjust R11 by a factor of the meas-

ured frequency divided by 100 kHz. This calculation is not entirely accurate, so check both frequencies after making the adjustment.

If the output of the schmitt trigger is connected to a spare bit of an input port you can have a go at software decoding of other recording formats such as Tarbell.

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